WALMART SALES PERFORMANCE DASHBOARD

Using Microsoft Power BI

PROJECT REPORT

Submitted By:

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Tool Used: Microsoft Power BI Desktop

1. Introduction

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps me convert raw data into meaningful reports and interactive dashboards.

For this project, I created a Walmart Sales Performance Dashboard using Microsoft Power BI. I used KPI cards, tables, charts and slicers to present sales information, compare store performance and study changes in sales over time.

The dashboard also helps me compare holiday and non-holiday sales and examine how total sales vary with fuel prices. By presenting the information visually, I can understand the sales data more easily.

2. Objectives of the Project

The main objectives of my project are:

- To analyse Walmart sales performance across 45 stores.
- To create an interactive dashboard using Microsoft Power BI.
- To compare sales performance between different stores.
- To identify changes in sales over the years.
- To compare holiday and non-holiday sales.
- To examine the relationship between fuel prices and total sales.
- To use KPI cards and slicers to make the report easier to analyse.

3. Dataset Description

I used Walmart sales data containing information about stores, sales, dates, fuel prices and holiday status. The report presents store-level values and yearly sales information for 2010, 2011 and 2012.

The main fields and measures used in my dashboard are:

Field	Description
Store	Identifies each Walmart store. The dashboard includes 45 stores.
Total Sales	Displays the total sales amount for a store or selected period.
Average	Displays the average value used for store comparison.
Transaction Count	Shows the number of transactions recorded.
Date / Year	Used to analyse sales over time and filter the report by date.
Fuel_Price	Used to compare fuel price values with total sales.
Holiday_Flag	Identifies holiday and non-holiday periods.

4. Data Preparation

Before creating the dashboard, I checked the data fields required for the report. Date and year fields need to be treated as date values, while sales, average and fuel-price fields need to be numeric so that Power BI can calculate and display them correctly.

I also used the store, date and holiday-status fields to organize the information for comparison and filtering. Correct data types and consistent field values are important because they help the charts and calculations display the intended results.

5. DAX Measures and KPI Cards

I used summary measures in the dashboard to display the main sales information. The KPI cards show the following values:

- **Total Sales:** approximately 6.74 billion.
- **Average:** approximately 1.05 million.
- **Store Count:** 45.
- **Transaction Count:** 6,437, displayed in rounded form as 6K.

The store-wise table also displays total sales, average and transaction count for individual stores. The total row shows total sales of approximately 6.7365 billion, an average value of 10,46,535.82 and 6,437 transactions.

6. Dashboard Components

I created different visualizations to understand the sales data from multiple perspectives.

6.1 Total Sales by Store

I used a column chart to compare total sales across the 45 stores. Each column represents one store. This chart helps me identify stores with higher and lower sales values.

Purpose: To compare store performance and identify differences in total sales.



Figure 1: Store-wise sales, average values and yearly sales trends.

6.2 Average by Store

I used a horizontal bar chart to compare the average value of each store. This layout makes it easier to view and compare many stores in one chart.

6.3 Weekly Sales Trend

I used an area chart to show total sales by year. Sales increase from approximately 2.3 billion in 2010 to around 2.45 billion in 2011. They then decrease to approximately 2.0 billion in 2012. This chart helps me identify the increase and decline in sales over the selected years.

6.4 Cumulative Weekly Sale Trend

I also used a line chart to display the cumulative weekly sales trend across 2010, 2011 and 2012. It provides another view of the change in sales over time.



Figure 2: Year-wise weekly and cumulative sales trends.

6.5 Holiday vs Non-Holiday Sales

I used a doughnut chart to compare sales for holiday and non-holiday periods. The chart displays approximately 6.23 billion (92.5%) for HOLIDAY and 0.51 billion (7.5%) for NON-HOLIDAY.

This chart helps me understand the share of total sales represented by each category. The percentages show each category's contribution to the displayed sales total; they do not by themselves compare average sales per week.

6.6 Total Sales by Fuel Price

I used a scatter chart to examine total sales in relation to fuel price. Each point represents a plotted fuel-price and sales value. The points do not show a clear straight-line pattern, so the chart does not indicate an obvious simple relationship between these two values.

6.7 Total Sales by Year and Store

I used a multi-series chart to compare total sales by year and store. The different coloured bands represent stores across 2010, 2011 and 2012. This helps me view how store contributions are distributed across the years.



Figure 3: Holiday sales, fuel-price comparison and sales by year and store.

7. Slicers Used

I added three filters to make the dashboard interactive:

- **Store:** I can select a particular store and view its sales information.
- **Date:** I can select a date to focus on the sales recorded during that period.
- **Holiday Flag:** I can filter the report by HOLIDAY or NON HOLIDAY.

These slicers help me analyse selected parts of the data. When I change a filter, the connected visuals update according to the selected value.

8. Key Findings

From my dashboard analysis, I identified the following findings:

- Total sales are approximately 6.74 billion.
- The dashboard includes 45 Walmart stores.
- The total transaction count is 6,437.
- The average KPI is approximately 1.05 million.
- Sales rise from 2010 to 2011 and decline in 2012 in the yearly trend charts.
- The HOLIDAY category represents approximately 92.5% of the displayed sales, while NON-HOLIDAY represents approximately 7.5%.
- The fuel-price scatter chart does not show an obvious simple linear pattern.
- Total sales and average values vary across stores.

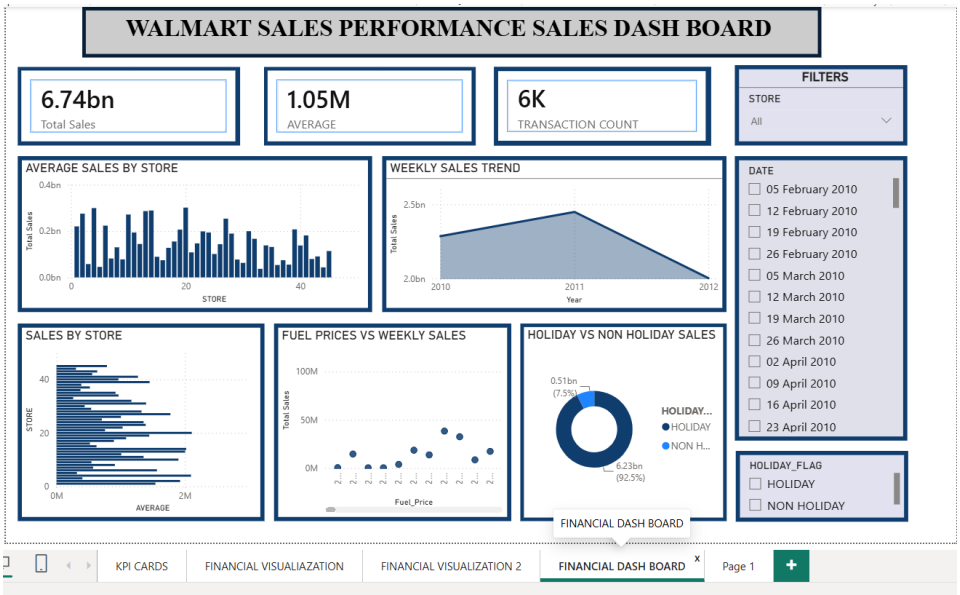


Figure 4: Walmart Sales Performance Dashboard.

9. Importance of the Dashboard

- **Easy Data Understanding:** I can understand sales information through charts and KPI cards.
- **Quick Analysis:** KPI cards provide the main sales figures in one place.
- **Store Comparison:** I can compare sales and average values across stores.
- **Trend Analysis:** Year-wise charts help me identify changes in sales over time.
- **Interactive Analysis:** Slicers allow me to focus on selected stores, dates and holiday categories.

10. Conclusion

Through this project, I learned how to use Microsoft Power BI to present sales data in the form of KPI cards, tables, charts and interactive slicers. I created a Walmart Sales Performance Dashboard to analyse sales across 45 stores and review important measures such as total sales, average and transaction count.

The visualizations helped me compare store performance, study yearly sales trends, examine holiday and non-holiday sales, and explore the relationship between fuel prices and sales. I also learned how slicers make a dashboard more interactive and allow me to focus on selected information.

Overall, this project improved my understanding of data visualization, dashboard creation and sales analysis using Power BI.

11. References

- Microsoft Power BI Desktop.
- Walmart sales dataset used for the project.
- Microsoft Power BI documentation.